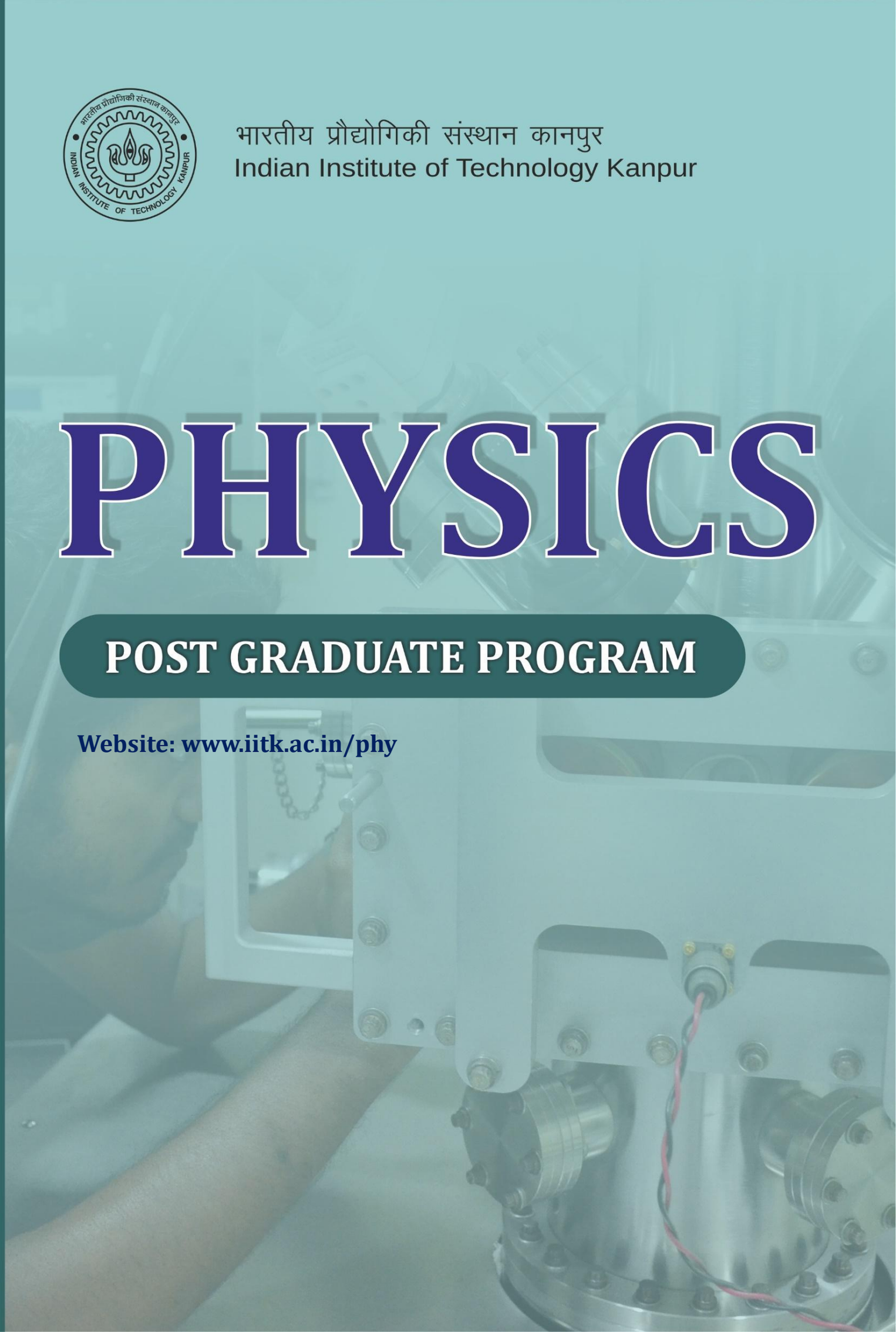
भारतीय प्रौद्योगिकी संस्थान कानपुर
Indian Institute of Technology Kanpur

# PHYSICS

## POST GRADUATE PROGRAM

**Website: www.iitk.ac.in/phy**



# PHD PROGRAM IN PHYSICS

The Department of Physics offers many subdisciplines in the PhD. program. The requirements are prescribed to ensure that the scholars acquire enough professional maturity to enable them to deal with a wide range of research problems in their respective fields of specialization as written below:

| | |
|---|---|
| 1. Complex System | Biological Fluids, Compartment Soft Matters, Exoplanet, Fluid Dynamics, Game Theory, Ion Beam, Non-linear Dynamics, Nuclear and Atomic Physics, Plasma Physics, Smart Surfaces, Soft and Active Matters, Turbulence, etc. |
| 2. Condensed Matter (Expt.) | 2D and Nano Materials, Magneto-electronics, Magneto-optics, Opto-spintronics, Quantum Materials, Quantum Devices, Strongly Correlated Electron Systems, Semiconductor Physics, STM, Superconductivity, Topological Materials, Ultrafast light-Matter Interaction, etc. |
| 3. Condensed Matter (Theory) | Density Functional Theory (DFT), Equilibrium and Non-Equilibrium Quantum Transport, Materials Informatics and Machine Learning (ML), Superconductivity and Magnetism, Ultra-cold Atoms, etc. |
| 4. High Energy Physics | AdS/CFT, Conformal Field Theory (CFT), Cosmology and Astrophysics, Experimental High Energy Physics, ML in HEP, Neutrino Physics, Quantum Field Theory (QFT), Quantum Chromodynamics (QCD), SM and BSM Phenomenology, String Theory, etc. |
| 5. Photonics and Quantum Optics | Biogenic Photonics, Cold Atoms and Quantum Metrology with Single Pure state of Cold Atoms and Ions, Fiber Optics, Laser Cooling and Trap, Light-Matter Interaction, Quantum Optics, Nonlinear Optics, Quantum Computation and Simulations with Cold Ions, etc. |

# ELECTIVE COURSES

- Advanced general relativity and black holes
- Advanced quantum and statistical mechanics
- Advanced quantum mechanics
- Advanced statistical mechanics
- Astrophysics
- Coherence & quantum entanglement
- Coherent optics
- Computational simulation methods in physics
- Concepts of plasma physics
- Condensed matter II
- Condensed matter phenomena in low dimensional materials
- Cooperative phenomena in classical & quantum systems
- Correlated electrons and quantum magnetism
- Cosmology
- Density functional theory
- Dynamical Systems and Chaos
- Electronic structure of materials
- Evolutionary game dynamics
- Gauge theory and renormalization
- General relativity & cosmology
- Green nanotechnologies
- Group theory and its applications to particle physics
- High Energy Physics II
- High Performance Computing with Applications
- Introduction group theory & its application to quantum mech.
- Introduction to astrophysical fluid
- Introduction to biophotonics
- Introduction to conformal field theory
- Introduction to high performance computing and machine learning
- Introduction to high performance computing for scientists and engineers
- Introduction to spintronics
- Introduction to x-ray neutron scattering
- Low temperature physics
- Machine Learning in Particle Physics
- Magnetism in materials
- Mathematical methods II
- Measurement techniques
- Metamaterials & plasmonics
- Non-equlibrium statistical mechanics
- Nonlinear system & dynamics
- Nuclear techniques in solid state studies
- Optical Imagining
- Particle physics
- Photonic devices

# ELECTIVE COURSES

- Physics and technology of thin films
- Physics at Hadron Collider
- Physics of life
- Physics of natural nano-machines
- Physics of semiconductor nanostructures
- Physics of turbulence
- Principles of lasers and detectors
- Quantum dynamics, information and computation
- Quantum dynamics: computation & information
- Quantum electronics
- Quantum field theory
- Quantum many body physics
- Quantum optics
- Quantum phases in nano-scale system
- Quantum Technology
- Soft matter physics
- Soft matter: concepts and methods
- Special & general relativity
- Special topics in quantum mechanics
- Stochastic dynamics of cognition
- Superconductivity and applications
- Techniques in Experimental High Energy Physics
- The standard model of fundamental physics
- Theory of random processes and applications
- Thermal physics at nano-scale
- Topics in semiconductor
- Transport in mesoscopic systems
- Turbulence in space plasmas
- Type II superconductors vortices & applications
- Uncertainty, Information, and Classical dynamics



# MAJOR RESEACH FACILITIES

**Condensed Matter Physics:**

Nitrogen and Helium Liquefiers, Superconducting magnets (up to 14 Tesla), Closed cycle Helium Refrigerators (down to 1.3 K), Experimental Setup for Resistivity, Hall Effect and Magnetic Susceptibility, Magnetoresistance, Specific Heat, Thermoelectric Power, Tunneling Conductance, Magnetostriction, Ultra high vacuum scanning probe microscope, superconducting quantum interference device (SQUID) based Magnetic Properties Measurement System (MPMS) with 10-6 emu resolution and 1.7K - 350K temperature range and magnetic field up to 5 Tesla. We also have a Physical Properties Measurements System (PPMS) with a 14 tesla superconducting magnet and helium–3 fridge with 300 mK base temperature. Scanning electron microscope (SEM) with electron beam lithography facilities for nano-scale patterning have been installed recently.

Some homemade facilities like variable temperature (8K - 300K) scanning tunneling microscope (STM) with atomic resolution and conductance imaging, magneto-optic Kerr effect (MOKE) imaging of magnetic surfaces, Pulsed Excimer Laser ablation facility for synthesis of magnetic, superconducting and dielectric superlattices, high frequency measurements of vortex dynamics in superconductors, Raman and microRaman Spectroscopy set-up for correlated systems. There are also experimental set ups for Dielectric measurement including Ferroelectric hysteresis of bulk and thin film materials, Thermal expansion measurement, Broadband FMR spectroscopy of ultrathin films, and Inverse Spin Hall Effect measurement.

Well-equipped facilities have been setup for opto-electronic characterization of semiconductor materials and devices. State of the art research facilities for organic semiconductors are available. The Department participates in major projects on Organic Electronics through Samtel Centre for display Technologies. For micro-fluidics, we have contact angle Goniometer, fluorescent microscope and high speed (10,000 fps) camera

**Condensed Matter Physics:**

ND:YAG lasers, Excimer Laser, Double Monochromators, Photon Counters, Spectrofluorimeter, Box Car Averagers, Vibration free tables. Facilities exist for High Resolution Spectroscopy, Micro-Raman Spectroscopy, Laser based pump-probe measurements, Laser-Plasma Studies and Nonlinear Optics, Diode Lasers, spectrograph-CCD system, Optical Spectrum Analyzers and Fibre Optics System, Optical Lithography Set-up, Atomic Force Microscopy Facility (Park XE 70 machine with a 100micron X 100 micron scanner head and a liquid cell). Two new Labs – Quantum and Nano-Optics, Quantum Optics and Entanglement Lab. are being set up.

# MAJOR RESEACH FACILITIES

**Ion Beam and Nuclear Techniques Laboratory:**

Mossbauer Spectrometer, Tandem particle accelerator, focused ion beam (FIB) with SEM column (Nova Lab 600) for Nano Microfabrication down to nm scales, modern 1.7 MeV Tandetron accelerator with capabilities of producing ion beams of almost all elemental species. The facility is equipped with a nuclear microprobe station for science and engineering applications such as proton and heavy ion beam writing, RBS, PIXE and ion Channeling E-beam deposition, Positron Annihilation, Mossbauer Studies at Low Temperatures.

**Computational Facilities:**

Physics cluster (Newton): 396 cores, 1.5 TB memory, Infiband switch. Chaos I: 256 cores, ½ TB RAM, Gigabit switch. Chaos II: 128 cores, ½ TB RAM, Two K10 GPU cards, Infiband switch. Many workstations including 3D capable visualisation stations. Dirac cluster for electronic structure calculations, 96 cores, 768 GB RAM. In addition, there are the following Institute facilities :hpc Cluster 1: 2994 cores, 100 TB storage, Peak rating 34.5 Tflops (Ranked 369 in June 2010 Top500 ranking). hpc Cluster II: 15,360 cores, 98TB RAM, Peak rating 316 Tflops (Ranked 130 in Nov 2013 Top500 ranking). The Department organizes a yearly “JagadishwarMahanty Distinguished Lecture” (since the year 2005) by an eminent scientist which provides unique opportunity of interaction with faculty and students, and a “Research Scholars Day” involving short talks and poster presentations by research scholars in the department apart from the weekly colloquium and seminars.

# FACULTY LIST

**Adhip Agarwala**, Ph.D. Indian Institute of Science, Bengaluru 2018.
**Areas of Interest:** Theoretical quantum condensed matter physics

**Aditya Kelkar**, Ph.D. Tata Institute of Fundamental Research, 2009.
**Area of research:** Experimental atomic collision physics, heavy ions and electron collisions, accelerator physics

**Anjan Kumar Gupta**, Ph.D. University of Kentucky, 2001
**Area of research:** Scanning Tunneling Microscopy, Superconductivity

**Amit Agarwal**, Ph.D. Indian Institute of Science, Bangalore, 2009
**Area of research:** quantum many-body effects, low-dimensional quantum systems

**Anand Kumar Jha**, Ph.D. University of Rochester, 2009
**Area of research:** Quantum-optics

**Arjun Bagchi**, Ph.D. Harish chandra Research Institute, 2009
**Area of research:** Theoretical High Energy Physics, String Theory

**Arijit Kundu**, Ph.D. Heinrich-Heine University, Germany, 2012
**Area of research:** Equilibrium and non-equilibrium quantum transport. Topological systems in condensed matter.

**Apratim Kaviraj**, Ph.D, IISc Bangalore, 2017
**Area of research:** Theoretical high energy physics, conformal field theories, critical phenomena

**Chanchal Sow**, Ph.D. Indian Institute of Science, Bangalore, 2014
**Area of research:** Magnetism in Complex Oxides, Interplay between Ferromagnetism and Superconductivity, Single Crystal and Thin Film Growth of Novel Quantum Materials, Strongly Correlated Electron Systems under Non Equilibrium Steady State (SCESness).

**Chandrima Banerjee**, Ph.D. S N Bose Center, 2017
**Area of research:** Experimental Condensed Matter Physics, Magneto-Optic Spectroscopy, Ultrafast Light Matter Interaction, XUV Spectroscopy

**Debtosh Chowdhury**, Ph.D. IISc, Bangalore, 2013
**Area of research:** Theoretical Particle Physics, Beyond the Standard Model phenomenology, Cosmology andAstroparticle physics

**Dipankar Chakrabarti** (HoD), Ph.D. Saha Institute of Nuclear Physics, 2004
**Area of research:** Quantum Chromodynamics, Lattice Gauge Theory, Lightcone Field Theories

**Diptarka Das**, Ph.D. University of Kentucky, 2014
**Area of research:** Conformal field theories, Non-equilibrium physics, String theory

**Gopal Hazra**, Ph.D. Indian Institute of Science,Bengaluru 2018
**Areas of Interest:** Astrophysical Magnetic fields, Exoplanets, Exoplanetary Atmospheres, Solar and Planetary winds, Space weather

**Goutam Das,** Ph.D Saha Institute of Nuclear Physics, 2017
**Areas of Interest:** Perturbative Quantum Chromodynamics, Effective Theories

# FACULTY LIST

**Harshwardhan Wanare**, Ph.D. University of Hyderabad, 1998
**Area of research:** Non-linear optics, imaging

**Jayita Nayak**, Ph.D. DeviAhilyaViswavidyalaya, Indore, 2015
**Area of research:** Angle resolved photoemission spectroscopy of topological insulators, high Tc superconductors, and hard X-ray photoelectron spectroscopy of quasicrystals and Heusler alloys

**Joydeep Chakrabortty**, Ph.D. Harishchandra Research Institute, 2011
**Area of research:** Particle Physics (Theory and Phenomenology); Main interest: Unification of fundamental forces, Supersymmetry, Supergravity

**Kaushik Bhattacharya**, Ph.D. Saha Institute of Nuclear Physics, 2005
**Area of research:** Cosmology, Particle Physics

**Krishnacharya**, Ph.D. Max Planck Institute, 2007
**Area of research:** Soft Matter Physics

**Koushik Pal**, Ph.D, JNCASR, 2017
**Area of research:** Materials theory and simulation, Materials informatics and machine learning, Quantum and energy materials

**Mahendra K Verma**, Ph.D. University of Maryland, 1994
**Area of research:** Nonlinear Physics, Turbulence, Nonequilibrium Statistical Physics, Astrophysics

**Manas Khan**, Ph.D. Indian Institute of Science, Bangalore, 2011
**Area of research:** Statistical Physics of Soft and Active Matter (Experiment, Modeling and Computation), Optical Trapping and Micromanipulation, Brownian dynamics simulation

**Manoj Kumar Harbola**, Ph.D. City University of New York, 1989
**Area of research:** Atomic and Molecular Physics

**Navaneeth Poonthottathil**, Ph.D., Cochin University of Science and Technology (CUSAT) and Fermilab, 2017
**Area of research:** Experimental High Energy Physics

**Nilay Kundu**, Ph.D. Tata Institute of Fundamental Research, 2014
**Area of research:** Theoretical High Energy Physics, String Theory

**R Vijaya**, Ph.D. IIT Madras, 1991
**Area of research:** Fiber optics, Nonlinear optics and Photonic band gap structures

**Rajeev Gupta**, Ph.D. Indian Institute of Science, 2001
**Area of research:** Optical spectroscopy and phases

**Rohit Medwal**, Ph.D., University of Delhi, 2014
**Areas of Interest:** Nanomagnetism, Spin transport and high frequency spin dynamics in spin devices, Terahertz spintronics

**Sagar Chakraborty**, Ph.D. S N Bose National Centre for Basic Sciences, 2009
**Area of research:** Nonlinear Dynamics, Evolutionary Game Theory

# FACULTY LIST

**Saikat Ghosh**, Ph.D. Cornell University, 2008
**Area of research:** Quantum Optics

**Satyajit Banerjee**, Ph.D. Tata Institute of Fundamental Research, 2000
**Area of research:** Superconductivity, Magneto-optics

**Sivasurendar Chandran**, Ph.D. Indian Institute of Science, Bangalore, 2014
**Area of research:** Polymer and Soft Matter Physics

**Soumik Mukhopadhyay**, Ph.D. Saha Institute of Nuclear Physics, 2009
**Area of research:** Spintronics, Nanomagnetism, Physics of complex oxide materials

**Sudeep Bhattacharjee**, Ph.D. RIKEN, Japan, 1999
**Area of research:** Plasma Physics, Focused ion beams from intense plasmas, Interaction of multi-element ion beams with matter, Physics of nano-scale systems

**Sudeep K Ghosh**, Ph.D., Indian Institute of Science Bangalore, 2017
**Areas of Interest:** Superconductivity and magnetism, Topological order, Ultracold atoms

**Sapam Ranjita Chanu**, Ph.D., Indian Institute of Science Bangalore, 2014
**Areas of Interest:** Quantum computation and simulations with cold ions, cold atoms and quantum metrology with single pure state of cold atoms and ions

**Sabyasachi Chakraborty**, Ph.D., IACS, Kolkata, 2016
**Areas of Interest:** Theoretical Particle Physics

**Sudipta Dubey**, Ph.D. Tata Institute of Fundamental Research, 2015
**Area of research:** Opto-electronics, Physics of atomically thin materials, Low-temperature electrical transport

**Supratik Banerjee**, Ph.D.Universite Paris-Sud, France 2014
**Area of research:** Turbulence in fluids, Physics of space and astrophysical plasmas

**Swagata Mukherjee**, Ph.D., University of Calcutta, 2016
**Area of research:** Experimental High Energy Physics, Collider experiments.

**Sanmay Ganguly**, Ph.D, TIFR, Mumbai, 2015
**Area of research:** Experimental High Energy Physics, Application of machine learning in HEP and other branches of physics

**Subhomoy Haldar**, Ph.D, Homi Bhabha National Institute, RRCAT, 2020
**Area of research:** Semiconductor Physics, Quantum Devices, Single Photon Detection, Magneto-Optical Transport

**Taraknath Mandal**, Ph.D., IISc, Bangalore, 2015
**Area of Research:** Computational Soft Matter Physics and Biophysics

**Tarun Kanti Ghosh**, Ph.D. Institute of Mathematical Sciences, 2003
**Area of research:** Utra-cold atomic gases, Nanoscopic physics

**Tapobrata Sarkar**, Ph.D. Institute of Mathematical Sciences, 2001
**Area of research:** String Theory & Quantum Gravity

**Venkata Jayasurya Yallapragada**, Ph.D., TIFR, 2017
**Area of research:** Nanophotonics, nanoscale light emitters and scatterers, biogenic photonic structures, optical imaging

**Zakir Hossain**, Ph.D. Tata Institute of Fundamental Research, 1997
**Area of research:** Superconductivity, Strongly-correlated electron systems

# BROAD RESEARCH AREAS

**The major research groups and active research areas are listed below:**

**Condensed Matter Experiments**

- Broadband FMR spectroscopy, spin pumping
- Charge Ordered State of Transition Metal Oxide Thin Films
- Defects and Disorder in Semiconductors
- Dielectric spectroscopy
- Effects of Localized States on Electrical and Optical Properties
- Electronic and Magnetic Properties of Transition Metal Oxides
- Electronic, Magnetic, Optical Properties of Surfaces using Scanning Probe Microscopy
- Hybrid Inorganic/Organic Devices
- Intense Laser-Matter Interaction
- Interplay between Magnetism and Superconductivity
- Low dimensional magnets
- Magnetic and Transport behaviour of Heavy Fermion Compounds
- Magneto-Optical Imaging
- Nanostructures of Superconducting and Magnetic Materials
- Non-Equilibrium Transitions in Driven Vortex States of Superconductors
- Quantum Devices
- Quantum Materials
- Printable Electronics, Organic LED and Lighting
- Soft Matter Physics
- Spin transport, Inverse Spin Hall Effect
- Spintronics, Spin-Polarized Tunneling in Transition Metal Oxides
- Strongly Correlated Electron Systems in Low Dimensions
- Ultrafast light-matter interaction

**Condensed Matter Theory**

- Antiferromagnetism, Superconductivity, BE Condensation and Disordered Systems
- Applications of Density Functional Theory
- Classical and Quantum Phase Transitions in Random Systems
- Correlated Electrons and Quantum Magnetism
- Electronic Structure of Atoms, Molecules, and Solids
- Electronic Structure of Disordered Systems
- Low Dimensional Electronic Systems, Graphene, Spintronics
- Molecular Dynamics Simulations and Genetic Algorithms
- Non-Equilibrium Dynamics of Magnetic and Non-Magnetic Systems
- Non-equilibrium Dynamics of Spin Systems
- One-Dimensional and Quasi-One Dimensional Spin and Electron Systems
- Phase Transitions in Driven Diffusive Systems
- Plasmonic Properties of Metallic Structures

# BROAD RESEARCH AREAS

- Quantum Entanglement and Quantum Computation
- Quantum Many Body Effects in Low Dimensional Systems
- Spin, Charge, Orbital Correlations in Transition Metal Oxides
- Structured Photonic Materials and Negative Refractive Materials
- Topological Insulators
- Transport Properties of Hybrid Structures

**Photonics and Quantum Optics**

- Bio-Medical Applications
- Coherent Control of Light-Matter Interaction
- Cold-Atom Systems, Foundations of Quantum Mechanics
- Imaging in Complex Media & Biological Tissues
- Laser Raman Spectroscopy
- Laser-Plasma Studies
- Nonlinear Optics, Fiber Optics
- Non-Linear Optics, Quantum Optics
- Photonic Band Gap Structures
- Photonics and Waves in Random Media
- Quantum Optics, Quantum Information, Precision Spectroscopy
- Using Cold-Atom Systems to Probe Nano-Materials and Molecules.

**High-Energy Physics**

- Astrophysics, Cosmology, Astroparticle Physics
- Black Holes Thermodynamics.
- Classical and Quantum Gravity,
- Experimental High Energy Physics (Neutrino Experiments and Collider Physics Experiments)
- Hydrodynamics from charged black branes, fluid dynamics and gravity
- Machine Learning in Particle Physics
- Particle Physics, Beyond Standard Model Phenomenology
- Quantum Chromodynamics, Lattice Gauge Theory, Effective Field Theory
- Quantum Field Theory
- String Theory, AdS-CFT and AdS-Condensed Matter correspondence.

# BROAD RESEARCH AREAS

## Plasma Physics and Ion beam Physics

- ❖ Atmospheric pressure low temperature micro-plasmas
- ❖ Atomically heterogeneous systems created with low energy ion beams
- ❖ Experimental plasma physics
- ❖ Ion-Beams for science, engineering and technology
- ❖ Physics of plasmas confined by a dipole magnet
- ❖ Plasma based multielement focused ion beams
- ❖ Research using focused ion beams
- ❖ Wave Interaction with plasmas confined in multi-cusp magnetic fields

## Complex Systems

- ❖ Analytical theory for different type of turbulent flows
- ❖ Astrophysics
- ❖ Biological and Statistical Physics
- ❖ Biological Fluids
- ❖ Brownian dynamics simulations
- ❖ Computational Soft Matter Physics and Biophysics
- ❖ Evolutionary Game Theory
- ❖ Exoplanets
- ❖ Experimental Plasma Physics
- ❖ Focused Ion Beams
- ❖ High Performance Computing
- ❖ Magnetohydrodynamics & Dynamo
- ❖ Nonequilibrium statistical physics
- ❖ Nonlinear Dynamics
- ❖ Optical Tweezers and Micromanipulations
- ❖ Origin of magnetic fields in the Sun and stars
- ❖ Physics of Nanoscale Systems
- ❖ Smart Surfaces with Tuneable Wetting Properties
- ❖ Soft Condensed Material
- ❖ Soft Condensed Matter
- ❖ Star-planet interaction and space weather
- ❖ Statistical physics of Soft and Active matters
- ❖ Surfaces and Interfaces
- ❖ Turbulence and Fluid Dynamics,
- ❖ Turbulence in living fluids

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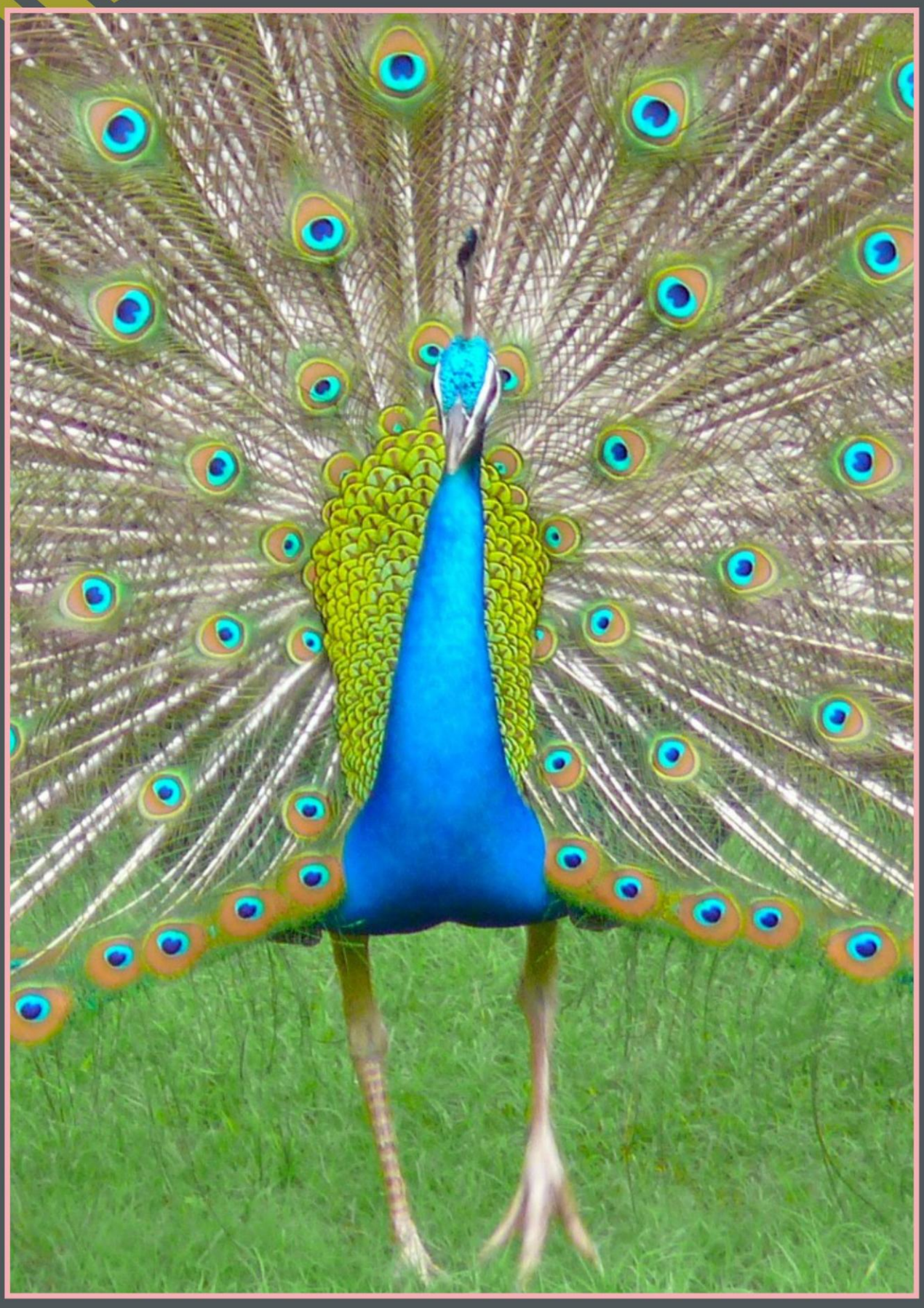
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